

Nguyen Duong Hieu

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Professional Summary

A dedicated Data Science student with a strong research interest in Causal Inference and its intersection with Machine Learning, particularly in causal representation learning, out-of-distribution (OOD) generalization, and trustworthy AI.

Education

National Economics University, BS in Data Science in Economics and Business Sept 2023 – Now

- GPA: 3.78/4.0
- **Coursework:** Linear Algebra, Probability & Statistics, Optimization, Machine Learning, Data Structures & Algorithms.

Tran Phu High School for the Gifted, Natural Science 1 2020 – 2023

Experience

Trainee, AI in Action Talent Training Program, Vingroup Summer 2026

- Selected for an intensive, hands-on training program focused on practical Artificial Intelligence applications.

Student Researcher, DAAI Lab – National Economics University, Hanoi, Vietnam Jan 2026 – Now

- Researched and developed innovative ML/AI algorithms applied to real-world enterprise problems, focusing on multimodal information extraction and intelligent business process automation, resulting in improved operational efficiency.
- Designed and implemented end-to-end AI pipelines integrating NLP, computer vision, and retrieval techniques to deliver production-ready solutions for enterprise clients.

Research Assistant, Project: “Does Network Position Cause Scholarly Success?” Spring 2026
with Prof. Mike Nguyen (University of Southern California) and Prof. Minh Nguyen (Florida Atlantic University)

- Curated and structured large-scale academic collaboration network datasets to support causal analysis of network position effects on scholarly productivity, enabling the team’s empirical investigation of structural causes of research outcomes.

Teaching Assistant (Tutor), Python for Data Science & Power Bi, National Economics University – Hanoi, Vietnam August 2025 – December 2025

- Supported students in Python and Power BI courses by explaining concepts, preparing assignments and teaching materials, and enhancing practical data science skills.
- Evaluated and graded student submissions, providing detailed and constructive feedback to improve understanding.

Student Researcher, DATAOPT Lab – National Economics University, Hanoi, Vietnam July 2024 – January 2026

- Researched and implemented mathematical optimization algorithms (e.g., convex optimization) and evaluated their practical performance in ML contexts through experiment reproduction.

Projects

Multimodal AI in Enterprise Robotic Process Automation Spring 2026

- Designed a modular two-stage **agentic workflow framework** integrating OCR and **ASR (PhoWhisper)** for transcription with a **zero-shot LLM (Gemini)** for structured entity extraction from unstructured business inputs.
- Developed a **hybrid retrieval mechanism** that combined dense semantic representations (Sentence-BERT) with sparse lexical matching (BM25) via Reciprocal Rank Fusion (RRF), achieving substantial improvements in retrieval accuracy.

- Achieved F1-scores of 0.91 (visual) and 0.85 (acoustic) under noisy conditions, maintaining a Recall@5 of 0.93 on a 50,000-SKU enterprise catalog.

Hanoi Temperature Forecasting System [GitHub]

Late 2025

- Developed an end-to-end ML pipeline to forecast Hanoi's daily temperature (5-day horizon) from 10+ years of data, engineering 90+ features (lag variables, STL decomposition, FFT) to capture temporal patterns and seasonality.
- Trained a multi-output CatBoost model with Optuna tuning ($R^2 \approx 0.84$); deployed a Streamlit app with SHAP explainability and an MLOps drift-detection pipeline (PSI, KS-test).
- *Technologies: Python, Pandas, Scikit-learn, CatBoost, Optuna, SHAP, ONNX, Streamlit*

Skills

Technical Skills

- Machine Learning & AI: Causal Inference (SCMs, DAGs, Invariant Prediction), Probabilistic Machine Learning, Large Language Models (LLMs), CatBoost, hyperparameter tuning (Optuna)
- Programming: Proficient in Python (Pandas, NumPy, Matplotlib), SQL; strong skills in Exploratory Data Analysis (EDA) and statistical analysis.
- Mathematics: Optimization, Probability & Statistics, Linear Algebra
- Tools: Power BI, Streamlit, LaTeX
- Domain Knowledge: Economics (microeconomics, macroeconomics fundamentals)

Publications

Promoting circular consumption behavior: Factors influencing Vietnamese Generation Z's intention to extend the lifespan of personal electronic devices

2026

Le Thu Thao, Nguyen Duong Hieu, Pham Phuong Trinh, Nguyen Minh Tung

22nd International Conference on Socio-economic and Environmental Issues in Development (ICSEED 2026)

- Integrated the Theory of Planned Behavior (TPB) and Push-Pull-Mooring (PPM) frameworks; employed PLS-SEM on 733 respondents to analyze determinants of electronic device lifespan extension among Vietnamese Gen Z.

Certifications and Achievements

- **2026** Third Prize, University-Level Scientific Research Competition, National Economics University
- **2025** Top 10, Hanoi Mathematical Modeling Competition
- **2024** Academic Encouragement Scholarship (Excellent), Faculty of Mathematical Economics, National Economics University
- **2024** Outstanding Youth Union Member Award, National Economics University
- **2023** Consolation Prize, Hai Phong City Mathematics Competition
- **2022** IELTS 7.0

Interests

- Causal Inference & Causal Machine Learning (Causal Discovery, Invariant Prediction)
- Causal Representation Learning & Out-of-Distribution (OOD) Generalization
- Trustworthy and Reliable AI Systems
- Probabilistic Machine Learning & Bayesian Methods
- Optimization and Applied Mathematics for Machine Learning